EXACTNESS OF THE SKYSCRAPER FUNCTOR AT A POINT OF A RINGED TOPOS, AND THE NON-LINEARITY OF ITS DIRECT SUM DECOMPOSITION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites three external items (one lemma, one section, one example), all recorded in assembly.external_references with canonical tags as navigation pointers.

Editorial note. The author's italicised 'not' in part (3) is part of the source text and was copied verbatim.

Allowed standard prerequisites (editorial).

- Exactness and the direct sum decomposition for the abelian-sheaves version. Stacks Project, Modules on Sites, Lemma lemma-skyscraper-exact (tag 09B1). Supplies parts (1) and the surjectivity of (2), and the decomposition whose linearity fails in (3).
- The topos of G-sets and its forgetful point. Stacks Project, Sites, Section sites-section-example-sheaf-G-sets (tag 00YR) and Example sites-example-point-G-sets (tag 00Z6). Provides the example on which the counterexample in (3) is built.

Source reference. Original lemma 05V7: sources/stacks-chow/tags.txt, source lines 7541–7541. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00VZ: sources/stacks-chow/tags.txt, source lines 1165–1165. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00YF: sources/stacks-chow/tags.txt, source lines 1253–1253. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(Sh(\mathcal{C}), \mathcal{O})$ be a ringed topos. Let p be a point of the topos $Sh(\mathcal{C})$.*

(1) *The functor $p_* : Mod(\mathcal{O}_p) \to Mod(\mathcal{O})$, $M \mapsto p_*M$ is exact.*
(2) *The canonical surjection $p^{-1}p_*M \to M$ is $\mathcal{O}_p$-linear.*
(3) *The functorial direct sum decomposition $p^{-1}p_*M = M \oplus I(M)$ of Lemma 05V7 is* **not** *$\mathcal{O}_p$-linear in general.*

Proof. Part (1) and surjectivity in (2) follow immediately from the corresponding result for abelian sheaves in Lemma 05V7. Since $p^{-1}\mathcal{O} = \mathcal{O}_p$ we have $p^{-1} = p^*$ and hence $p^{-1}p_*M \to M$ is the same as the counit $p^*p_*M \to M$ of the adjunction for modules, whence linear.

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Proof of (3). Suppose that G is a group. Consider the topos $G\text{-}Sets = Sh(\mathcal{T}_G)$ and the point $p : Sets \to G\text{-}Sets$. See Sites, Section 00VZ and Example 00YF. Here p^{-1} is the functor forgetting about the G-action. And p_* is the right adjoint of the forgetful functor, sending M to $\mathrm{Map}(G, M)$. The maps in the direct sum decomposition are the maps

$$M \to \mathrm{Map}(G, M) \to M$$

where the first sends $m \in M$ to the constant map with value m and where the second map is evaluation at the identity element 1 of G. Next, suppose that R is a ring endowed with an action of G. This determines a sheaf of rings $\mathcal{O}$ on $\mathcal{T}_G$. The category of $\mathcal{O}$-modules is the category of R-modules M endowed with an action of G compatible with the action on R. The R-module structure on $\mathrm{Map}(G, M)$ is given by

$$(rf)(\sigma) = \sigma(r)f(\sigma)$$

for $r \in R$ and $f \in \mathrm{Map}(G, M)$. This is true because it is the unique G-invariant R-module structure compatible with evaluation at 1. The reader observes that in general the image of $M \to \mathrm{Map}(G, M)$ is not an R-submodule (for example take $M = R$ and assume the G-action is nontrivial), which concludes the proof. $\square$